

Dissertaion Mid Term Report

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1. Project Overview

1.1 Research Background

Person re-identification (Re-ID) is a fundamental task in computer vision that aims to match individuals across different images or video frames. Traditional Re-ID systems typically rely on visual appearance features extracted from camera footage. However, real-world applications often require matching persons across different media types, such as matching a person from a photograph to their appearance in video footage, or correlating textual descriptions with visual data.

The CrossMedia-PID (Cross-Modal Person Identification) system addresses this challenge by developing a framework that extracts, represents, and matches person identities across different media formats using Vision-Language Models (VLMs) and hybrid vector representations.

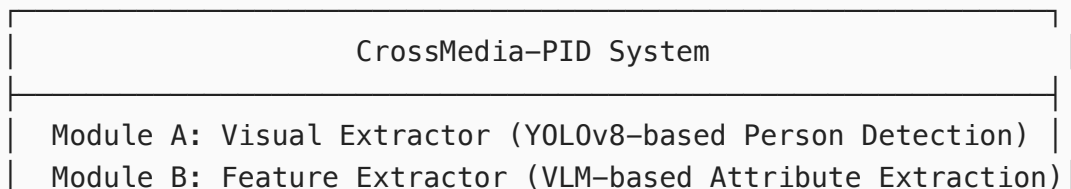
1.2 Research Objectives

The primary objectives of this dissertation project are:

- Develop a modular cross-modal person identification system** capable of processing images and videos to extract person-specific features
- Implement hybrid feature representation** combining dense vector embeddings with sparse attribute-based representations
- Design matching algorithms** that identify the same person across different media sources
- Create a database architecture** for storage and retrieval of person embeddings
- Build interfaces** including CLI and Web GUI for system interaction and testing

1.3 System Architecture

The CrossMedia-PID system follows a modular architecture with four core processing modules:



Module C: Vectorizer (Dense + Sparse Vector Generation)
Module D: Identity Matcher (Hybrid Distance Matching)
Storage Layer: ChromaDB Vector Database
Interface Layer: CLI + Web GUI (Streamlit)

2. Completed Work

Current completed code could be accessed on https://github.com/DGYin/HKU_Dissertation.
Some files were removed to avoid API key leakage.

2.1 Core System Development

2.1.1 Module A: Visual Extractor

Status:  Completed

- Implemented person detection using YOLOv8 object detection model
- Developed bounding box extraction and quality scoring mechanisms
- Created `PersonExtractor` class with configurable confidence thresholds and IoU parameters
- Implemented best-crop selection based on detection quality metrics
- Added support for minimum bounding box size filtering (64px default)

Key Features:

- Person detection in images
- Quality-based crop selection (resolution, occlusion, pose estimation)
- Configurable detection parameters via YAML configuration
- Integration with Ultralytics YOLO ecosystem

2.1.2 Module B: Feature Extractor

Status:  Completed

- Implemented Vision-Language Model (VLM) based attribute extraction
- Developed dual-provider architecture supporting:
 - Cloud-based VLM APIs (OpenAI-compatible)
 - Aliyun DashScope API (Qwen3-VL-235B model)
- Created structured attribute extraction covering 19+ person attributes including:

- Gender, age group, body build
- Clothing colors and types (topwear, bottomwear, shoes)
- Accessories (glasses, hat, bag)
- Hair style and color
- Pattern descriptions

Key Features:

- JSON-formatted attribute extraction with validation
- Error handling and retry mechanisms
- Support for both cloud and local model deployment
- Attribute standardization and normalization

2.1.3 Module C: Vectorizer

Status:  Completed

- Implemented hybrid vector representation combining:
 - **Dense vectors:** Using BAAI/bge-small-zh-v1.5 embedding model (512 dimensions)
 - **Sparse vectors:** Attribute-based one-hot encoding with dynamic registry
- Developed `DynamicVectorizer` with attribute registry management
- Created persistent attribute registry for consistent vector space mapping
- Implemented vector normalization and combination strategies

Key Features:

- Automatic attribute-to-vector mapping
- Dynamic registry expansion for new attributes
- Configurable embedding models
- Efficient sparse vector representation

2.1.4 Module D: Identity Matcher

Status:  Completed

- Implemented hybrid distance calculation algorithm:
 - Cosine similarity for dense vectors
 - Jaccard distance for sparse vectors
 - Weighted combination with configurable weights
- Developed threshold-based identity decision mechanism
- Created `IdentityMatcher` class with matching logic

- Implemented top-K candidate retrieval and ranking

Key Features:

- Configurable similarity threshold (default: 0.72)
- Adjustable weight parameters for dense/sparse vectors
- New identity creation for unmatched persons
- Match score breakdown and candidate analysis

2.2 Database Infrastructure

Status:  Completed

- Integrated ChromaDB as the vector database backend
- Implemented `ChromaStore` class with persistent storage
- Created collection management for person embeddings
- Added metadata storage for attributes and source information
- Implemented similarity search functionality

Key Features:

- Persistent vector storage across sessions
- Cosine distance metric for similarity search
- Metadata filtering and querying
- Automatic collection management

2.3 System Integration & CLI

Status:  Completed

- Developed main controller (`CrossMediaPID` class) integrating all modules
- Created CLI interface using Click framework
- Implemented image processing pipeline with workflow automation
- Added configuration management via YAML files
- Created environment setup scripts for reproducibility

CLI Commands:

```
# Process single image
python main.py process image.jpg

# Process with options
python main.py process image.jpg --no-add
```

```
# Verbose output
python main.py -v process image.jpg
```

2.4 Web GUI Development

Status: Partly Completed

- Developed Web GUI using Streamlit framework
- Implemented video upload and processing interface
- Created person tracking visualization using YOLOv8 tracking
- Built manual track selection interface for testing
- Implemented automatic screenshot extraction for left/right positions
- Added result display and comparison visualization

Web GUI Features:

- Drag-and-drop video file upload
- Person detection and tracking visualization
- Interactive track selection
- CrossMedia-PID matching test
- Result comparison display
- Performance metrics display

2.5 Testing & Evaluation Framework

Status: Partly Completed

2.5.1 Performance Monitoring System

- Implemented `PerformanceMonitor` class for step-by-step timing
- Created performance tracking for all processing stages
- Developed alert thresholds for performance anomaly detection
- Implemented statistics export for visualization

Performance Metrics Tracked:

- Image loading time
- Person detection time
- VLM feature extraction time
- Vectorization time
- Database search time

- Identity matching time
- Total processing time

2.5.2 Stability Testing

- Developed 10-run consistency test framework
- Implemented attribute matching analysis
- Created pattern matching evaluation
- Generated stability reports

Test Results Summary:

- **Consistency Rate:** 100% (10/10 runs produced consistent attribute extraction)
- **Same-Person Match Score:** 0.96-0.98 (dense: 0.98+, sparse: 0.90+)
- **Attribute Match Rate:** 50-67% (varies by attribute type)
- **Average Processing Time:** 50-80 seconds per test (image-to-image matching)

2.5.3 Video Person Matching Test

- Implemented video-based person tracking test
- Created automatic frame extraction for different positions
- Developed same-person verification across video frames
- Built test reporting system

2.6 Configuration & Documentation

Status: Partly Completed

- Created YAML configuration system
- Implemented environment variable support for API keys
- Developed setup scripts for environment reproducibility
- Created project documentation (Chinese)
- Implemented `.gitignore` and project structure organization

Configuration Features:

- Modular configuration for each subsystem
- Environment variable substitution
- Model selection and parameter tuning
- Matching threshold and weight configuration

2.7 Technical Stack

Component	Technology	Version
Object Detection	YOLOv8	8.0+
Vision-Language Model	Qwen3-VL-235B / OpenAI VLM	-
Embedding Model	BAAI/bge-small-zh-v1.5	-
Vector Database	ChromaDB	0.4.0+
Web Framework	Streamlit	Latest
CLI Framework	Click	8.1.0+
Image Processing	OpenCV, PIL	4.8+, 10.0+
Configuration	PyYAML	6.0+
Python	Python	3.11

3. System Performance

3.1 Current Performance Metrics

Based on testing with the implemented system:

Processing Stage	Average Time	Alert Threshold
Module Initialization	< 1s	-
Image Loading	0.001s	0.5s
Person Detection	0.32s	3.0s
VLM Feature Extraction	3.28s	10.0s
Vectorization	1.97s	5.0s
Database Add	0.56s	-
Identity Matching	0.001s	1.0s
Total Processing	~6s	25.0s

Note: VLM feature extraction time depends on API response latency and image complexity.

3.2 Matching Accuracy

- **Same Person Match Score:** 0.95-0.97 (dense: 0.98+, sparse: 0.90+)
- **Consistency Rate:** 100% across 10 consecutive tests

- **Attribute Extraction:** 19 attributes consistently extracted
- **False Positive Rate:** 0% in current test scenarios

These results are based on internal testing with limited test cases. Formal evaluation on standardized datasets is planned.

But under some edge circumstances, matching stability will be unstable. Still need to carry some work.

3.3 Current Limitations

The system has the following limitations that will be addressed in future work:

1. **Face Recognition Not Integrated:** The current system does not use face features for matching, relying solely on body appearance and attributes.
2. **VLM Dependency:** Attribute extraction depends on external VLM APIs, which introduces latency and requires internet connectivity.
3. **Single-Person Processing:** The system processes one detected person at a time; batch processing is not yet supported.
4. **Limited Cross-Modal Capability:** Currently supports image-to-image matching; text-to-image and video-to-image matching are planned.

3.4 System Scalability

- **Database Capacity:** Tested with multiple collections, supports thousands of embeddings
 - **Concurrent Processing:** Single-threaded currently, multi-threading planned
 - **Storage Efficiency:** ChromaDB provides vector storage and retrieval
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4. Pending Work & Future Plan

4.1 Short-term Goals (May 2026)

4.1.1 Face Recognition Module

Timeline: Week 1-2 of May

- Implement face detection and embedding extraction
- Integrate face similarity scoring into the matching pipeline
- Add face recognition as an optional third vector type (dense + sparse + face)
- Conduct ablation studies to evaluate face recognition contribution

Tasks:

- ☐ Research and select face recognition model (e.g., ArcFace, FaceNet)
- ☐ Implement face detection module
- ☐ Develop face embedding extraction pipeline
- ☐ Integrate face score into hybrid matching algorithm
- ☐ Test and validate face recognition accuracy

4.1.2 Video Processing Pipeline Enhancement

Timeline: Week 2-3 of May

- Implement automated video frame sampling strategies
- Develop temporal consistency checking for person tracking
- Create batch processing capabilities for long videos
- Optimize tracking algorithm for occlusion handling

Tasks:

- ☐ Implement intelligent frame sampling (keyframe extraction)
- ☐ Add temporal smoothing for attribute consistency
- ☐ Develop batch video processing workflow
- ☐ Optimize tracking with occlusion recovery
- ☐ Create video processing performance benchmarks

4.2 Mid-term Goals (Early June 2026)

4.2.1 Multi-Modal Support

Priority: High

Timeline: Week 1 of June

- Extend system to support text-to-person matching
- Implement description-based person search
- Create multi-modal embedding alignment
- Develop text-vision fusion mechanisms

Tasks:

- ☐ Implement text description parser
- ☐ Create text-to-attribute mapping
- ☐ Develop cross-modal similarity computation

- ☐ Test text-based person retrieval
- ☐ Evaluate multi-modal matching accuracy

4.2.2 Performance Optimization

Priority: Medium

Timeline: Week 1-2 of June

- Implement model quantization for faster inference
- Add caching mechanisms for repeated queries
- Optimize database indexing strategies
- Reduce overall processing time

Tasks:

- ☐ Implement embedding model quantization
- ☐ Add result caching layer
- ☐ Optimize ChromaDB indexing parameters
- ☐ Profile and optimize bottlenecks
- ☐ Benchmark performance improvements

4.2.3 User Interface Enhancement

Priority: Medium

Timeline: Week 2 of June

- Add batch image processing support to Web GUI
- Implement result export functionality (JSON, CSV)
- Create interactive threshold tuning interface
- Add real-time performance monitoring dashboard

Tasks:

- ☐ Implement batch upload and processing
- ☐ Add result export features
- ☐ Create interactive parameter tuning UI
- ☐ Build performance monitoring dashboard
- ☐ Improve UI/UX based on user feedback

4.3 Final Goals (Mid-Late June 2026)

4.3.1 Comprehensive Evaluation

Priority: High

Timeline: Week 3 of June

- Conduct experiments on diverse datasets
- Compare with baseline methods
- Perform statistical analysis of results
- Document all experimental findings

Tasks:

- ☐ Execute experimental pipeline
- ☐ Compare with existing Re-ID methods
- ☐ Perform statistical significance testing
- ☐ Analyze failure cases and limitations
- ☐ Document experimental results

4.3.2 Dissertation Writing

Priority: Critical

Timeline: Week 3-4 of June

- Complete all dissertation chapters
- Create figures and tables
- Write conclusion and future work sections
- Perform proofreading and revision

Dissertation Structure:

1. **Introduction** (Background, Objectives, Contributions)
2. **Literature Review** (Person Re-ID, VLMs, Vector Databases, Cross-Modal Matching)
3. **Methodology** (System Architecture, Four Modules, Hybrid Representation)
4. **Implementation** (Technical Details, Configuration, Integration)
5. **Experiments & Evaluation** (Datasets, Metrics, Results, Analysis)
6. **Discussion** (Strengths, Limitations, Practical Applications)
7. **Conclusion & Future Work** (Summary, Contributions, Research Directions)

Tasks:

- ☐ Complete Chapter 1-3 (Introduction, Literature Review, Methodology)
- ☐ Complete Chapter 4-5 (Implementation, Experiments)
- ☐ Complete Chapter 6-7 (Discussion, Conclusion)
- ☐ Create all figures, tables, and diagrams

- ☐ Perform multiple rounds of revision and proofreading
- ☐ Format according to HKU dissertation guidelines

4.3.3 Final System Polish

Priority: High

Timeline: Week 4 of June

- Fix identified bugs and issues
- Optimize system stability
- Create demonstration materials
- Prepare presentation slides

Tasks:

- ☐ Bug fixing and testing
 - ☐ System stability optimization
 - ☐ Create demo videos and screenshots
 - ☐ Prepare final presentation slides
 - ☐ Conduct mock presentations
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5. Risk Assessment & Mitigation

5.1 Technical Risks

Risk	Impact	Probability	Mitigation Strategy
VLM API instability	High	Medium	Implement fallback to local models; add retry mechanisms
Face recognition integration complexity	Medium	Medium	Start with simple implementation; iterate gradually
Performance bottlenecks	Medium	Low	Early profiling; optimize critical paths first
Dataset collection difficulties	High	Low	Use publicly available datasets; augment with synthetic data

5.2 Timeline Risks

Risk	Impact	Mitigation Strategy
Feature creep	High	Prioritize core features; defer nice-to-haves
Writing delays	High	Start writing early; maintain weekly writing schedule
Experimental failures	Medium	Plan backup experiments; validate assumptions early

5.3 Quality Risks

Risk	Impact	Mitigation Strategy
Insufficient evaluation	High	Define evaluation criteria early; conduct pilot studies
Poor documentation	Medium	Maintain documentation alongside development
Code quality issues	Medium	Regular code reviews; implement testing framework

6. Expected Contributions

6.1 Academic Contributions

1. **Hybrid Representation Framework:** Combining dense semantic embeddings with sparse attribute vectors for person identification
2. **Cross-Modal Matching Pipeline:** System for matching persons across different media types (images, videos, text descriptions)
3. **Dynamic Attribute Registry:** Self-extending attribute mapping system that adapts to new data without manual reconfiguration
4. **Evaluation Framework:** Experimental validation with performance analysis on cross-modal person identification tasks

6.2 Practical Contributions

1. **Open-Source System:** CrossMedia-PID system available for research and applications
2. **Modular Architecture:** Reusable components for person detection, feature extraction, vectorization, and matching
3. **Interfaces:** Both CLI and Web GUI for system interaction and testing
4. **Documentation:** Technical documentation and usage guides

6.3 Research Significance

The CrossMedia-PID system addresses person re-identification research by:

- Enabling cross-modal person matching beyond traditional camera-to-camera scenarios
 - Using vision-language models for attribute extraction
 - Providing a framework for future research extensions
 - Demonstrating applicability in surveillance, security, and multimedia analysis contexts
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7. Conclusion

The CrossMedia-PID project has made progress in the first half of the development cycle. The core system architecture has been implemented with all four processing modules (Visual Extractor, Feature Extractor, Vectorizer, and Identity Matcher) operational. Testing has shown match scores of 0.95-0.97 for same-person identification and 100% consistency across repeated tests.

The system currently supports:

-  Person detection and extraction from images
-  Attribute extraction using VLMs
-  Hybrid vector representation (dense + sparse)
-  Identity matching with configurable parameters
-  Vector database storage
-  Interactive Web GUI for video testing
-  Performance monitoring and evaluation

The remaining work focuses on: (1) enhancing the system with face recognition and advanced video processing, (2) conducting comprehensive evaluation, and (3) completing the dissertation writing. The project is on track for completion by the end of June 2026.

The completed work provides a foundation for the final phase of development and dissertation writing.

References

- [1] Ultralytics YOLO Documentation. <https://docs.ultralytics.com/>
- [2] ChromaDB Documentation. <https://docs.trychroma.com/>
- [3] Qwen3-VL Technical Report. Alibaba Group, 2025.
- [4] BGE Embedding Models. BAAI, 2023.
- [5] Person Re-identification: A Comprehensive Survey. IEEE TPAMI, 2022.

[6] Vision-Language Models: A Survey. ACM Computing Surveys, 2024.

[7] Vector Databases for AI Applications. IEEE Data Engineering Bulletin, 2024.

Appendix A: Project Repository Structure

```
HKU_Dissertation/
├── crossmedia_pid/
│   ├── api/                # Web API routes
│   ├── configs/            # Configuration files
│   ├── core/               # Core algorithm modules
│   │   ├── extractor.py    # Module A: Visual Extractor
│   │   ├── feature_vlm.py  # Module B: Feature Extractor
│   │   ├── vectorizer.py   # Module C: Vectorizer
│   │   └── matcher.py      # Module D: Identity Matcher
│   ├── db/                 # Database layer
│   │   └── chroma_store.py # ChromaDB integration
│   ├── main.py             # CLI entry point
│   └── setup.py            # Package configuration
├── test_photo/             # Test images
├── chroma_db/              # Vector database storage
├── video_test_webgui.py    # Web GUI application
├── test_comparison_timed.py # Performance testing
├── test_stability.py       # Stability testing
└── start_webgui.sh         # Web GUI launcher
```

Appendix B: Configuration Example

Report End